

Growth beats Value: Smart Beta, Factor Timing and Momentum in Global Equity Strategies

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Abstract

We investigate the relationship between Value, Growth and two forms of Momentum across a wide range of developed and emerging international equity markets using MSCI total return 'smart beta' indices. As would be anticipated, Value generally beats Growth. A distinction is then made between **relative momentum** where assets are ranked according to their performance against each other, and **absolute momentum** (sometimes known as trend following) where assets are categorized according to whether they have recently exhibited positive nominal return characteristics. We find that both Value and Growth portfolios benefit from momentum filters but particularly the latter, though they do suffer from familiar equity market volatility, skewness and substantial drawdowns. When we overlay simple absolute momentum on these factor indices, it typically delivers a much more favourable investment performance than relative momentum with considerably lower volatility and smaller drawdowns. Well-known behavioural finance biases help us understand such behaviour.

1. Introduction

Smart beta equity investing strategies are receiving more and more attention as alternatives to both stock-picking and long-only, market capitalisation weighted, 'passive' alternatives. While the holy grail of successful market timing with switching/'rotating' between factors is still considered elusive, some recent research suggests progress (see Alenc et al, 2015 and Brightman et al, 2017). For example, Brightman et al show that in order to maximize risk-adjusted returns whilst diversifying across some popular smart beta strategies, systematic rebalancing to fixed weights for individual factors is required. This reduces exposure to factors that have outperformed over recent years whilst increasing exposure to the less successful factors that have underperformed. Even this simple strategy is found to improve performance relative to a buy-and-hold weighting. Further refinement of dynamic rebalancing of factor exposures using short-term momentum and long-term reversal signals improves the return even more.

In this paper we revisit the 'Value versus Growth' debate using international MSCI country style indices in the context of absolute and relative momentum strategies and show how these techniques can not only enhance portfolio performance but also shed new light on a key component of many smart-beta strategies.

The outperformance of Value stocks over Growth stocks has been well documented in the finance literature. Fama and French (1992) demonstrate that sorting firms according to their book-to-market ratio has the ability to explain future returns. Other value metrics for individual stocks have included dividend yield (Christie, 1990), total payout yield (Boudoukh et al, 2007) and price-to-cash flow (Lakonishok et al, 1994). Greenblatt (2010) shows that a combination of an earnings-based value measure and return on equity has found stocks that have historically outperformed the broader market. Furthermore, value can be applied at the aggregate level, with Shiller's (2005) cyclically-adjusted price-earnings ratio being one of the best known examples. Whilst the former have focussed on the US equity market, Fama and French (1998) and Asness et al (2013), amongst others, have shown that value strategies work in international markets. Recently, Folkenshteyn, D, et al (2017)) found that value stocks outperform both market averages and growth stocks during market declines for 5 international markets.

The reasons why value strategies have been favourable to investors has been the source of much debate. Candidate explanations include that the excess returns to value are attributable to investors bearing some additional risk and thus they are merely receiving the appropriate compensation; e.g., for distress risk (Fama and French, 1992). Alternatively, Lakonishok et al (1994) propose a behavioural interpretation with growth stocks underperforming because investors extrapolate past earnings growth too far forward thus leading to over-optimism and excessively high valuations.

However, such conclusions are still not universally accepted. Phalippou (2008) suggests that size and institutional ownership may actually be at work influencing the result, while Houge and Loughran (2006) do not find value outperforming growth for either US equity style indices, large-cap firms or equity mutual funds. Scislaw and McMillan (2012) compare popular style index construction created by commercial providers with the Fama-French definitions of 'value' and find no outperformance for the former. Similarly, Chan et al (2002) consider alternative approaches to capturing value and suggest that after adjusting for style, growth managers actually outperform value managers.

In this paper we examine the performance of Growth and Value in the context of MSCI style indices for both developed (since 1974) and emerging (since 1987) stock markets. Contrary to Houge and Loughran (2006) and Scislaw and McMillan (2012), we see that the traditional finding that Value outperforms Growth is clearly present in our MSCI style index data. However, when we overlay a simple absolute momentum/ trend following on the individual country returns we see much more similar returns for Growth and Value portfolios. Yet, when we apply a relative momentum (Moskowitz et al, 2012) portfolio formation process across the universe of individual equity indices, Growth beats Value easily though volatility and drawdown is still elevated. Overlaying trend following on relative momentum leads to even greater outperformance of Growth over Value, with reduced skewness. In other words, reducing tail risk seems to enhance the performance of Growth investing. One explanation for this result relates to the response of the strategy returns to funding liquidity risk.

Of course, both relative and absolute momentum are examples of active management in terms of switching between assets and between assets and cash in a systematic, rule-driven way. Hence, even rather crude factor-timing procedures can lead to substantial enhancement of smart-beta equity strategies.

To anticipate our findings:

- Relative momentum strategies work across both Value and Growth stocks but the effect is larger in the latter.
- In developed markets, despite Value outperforming Growth on a buy-and-hold basis, a relative growth momentum strategy outperforms a relative value momentum strategy.
- Trend following/absolute momentum provides substantial improvements to risk adjusted returns in both Value and Growth.
- Trend following overlaying relative momentum('dual momentum') displays superior performance compared to relative momentum alone across all portfolios-and 'dynamic' Growth dominates 'dynamic' Value portfolios

2. Relative and Absolute Momentum

Relative Momentum

Momentum is one anomaly in the financial literature that has been demonstrated to offer some explanatory ability for future returns. Many researchers, such as Jegadeesh and Titman (1993) and Grinblatt and Moskowitz (2004), have focused on momentum at the individual stock level, while others, such as Miffre and Rallis (2007) and Erb and Harvey (2006), have observed the effect in commodity futures. Asness, Moskowitz, and Pedersen (2013) find momentum effects within a wide variety of asset classes. Typical momentum strategies involve ranking assets based on their past return (often the previous 12 months) and then buying the winners and selling the losers. This is known as 'relative' or 'cross-section' momentum. Ilmanen (2011) argues that this is not the ideal approach and that investors would be better served by volatility weighting the past returns. Failing to do this leads to the most volatile assets spending a disproportionate amount of time in the highest and lowest momentum portfolios (see also, Asness, Moskowitz and Pedersen, 2013).

Trend Following /Absolute Momentum

Trend following has been widely used in futures markets, particularly commodities, for many decades (see Ostgaard, 2008). Trading signals can be generated by a variety of methods, such as moving average crossovers and breakouts with the aim to determine the trend in prices. Long positions are adopted when the trend is positive, and short positions, or cash, are taken when the trend is negative. As trend following is generally rules based, it can aid investors because losses are mechanically cut short and winners left to run. This is frequently the reverse of investors' natural instincts. The return on cash is also an important factor either as collateral in futures or as the risk-off asset for long-only methods. Examples of studies on the effectiveness of trend following are those by, among others, Szakmary, Shen, and Sharma (2010) for commodity futures, and ap Gwilym et al. (2010) for equity indexes. Faber (2007) uses trend following as a means of tactical asset allocation and demonstrated that it is possible to form a portfolio that has equity-level returns with bond-level volatility. Ilmanen (2011) offers a variety of explanations as to why trend following may have been successful historically, including investor underreaction to news and herding behaviour. Moskowitz, Ooi, and Pedersen (2012) refer to an equivalent of trend following as "*time-series momentum*." They find significant time-series momentum in equity index, currency, commodity and bond futures for each of the 58 liquid instruments considered. They find persistence in returns for 1 to 12 months that partially reverses over longer horizons, consistent with sentiment theories of initial underreaction and delayed overreaction.

Combining Relative Momentum and Trend Following

A few studies have sought to combine some of the strategies discussed above. Faber (2010) uses momentum and trend following in equity sector investing in the United States. Antonacci (2012) uses momentum for trading between pairs of investments and then applied a quasi-trend-following filter to ensure that the winners have exhibited positive returns. The risk-adjusted performance of these approaches has been a significant improvement on benchmark buy-and-hold portfolios. In a related study, these ideas are extended to the multi-asset context (Clare et al, 2016) with the finding that although adding a relative momentum filter increases the level of return compared with equal weighting, momentum portfolios are prone to large drawdowns.

The evidence presented by Clare et al (2016) for a variety of asset classes, suggests that when cross-sectional momentum is compared to trend following, it is the latter that proves the more effective investment strategy. Risk-adjusted returns are higher with maximum drawdowns substantially reduced. The combination of the two momentums, e.g. 'winners' that are in an upward trend, offers much higher nominal levels of return compared to trend following alone but volatility also rises considerably too.

The foundations of theoretical support for understanding trend following and similar concepts lies in our understanding of behavioural finance with psychological insights into herding, regret and conservatism. These are reviewed in the survey of prospect theory by Barberis (2013). In this context investors can overcome such biases which destroy investment returns (see Dalbar, 2015) by using rigid quantitative investment rules which take discretion away from investors and reflect what we know about investor preferences for risk and return. The Dalbar study estimates that only about 15% of investors want to 'beat the market' but twice that percentage show extreme loss aversion: so how can we design investments (and investment strategies) that will avoid such emotional responses as 'bailing out' too early? The role of behavioural finance in understanding trend following and relative momentum is explored further in Clare et al (2016).

3. Value, Momentum and Protecting Against Drawdowns

Whilst Value and Momentum are well established investment strategies, there has been less attention focused on the interaction between the two. For a stock to become cheap on a value metric it needs the fundamentals to improve relative to the price. In the case of an earnings yield (E/P) approach, this means that E needs to increase (decrease) at a faster (slower) rate than P rises (falls). In the cases where P falls this makes it less likely the stock will qualify as a cross-sectional momentum winner and almost certainly rules it out from having positive absolute momentum. That aside though, within any group of value stocks there will be winners and losers, and when value stocks rise they will have upward trends.

Asness (1997) studies the interaction between value (both book to market B/M and dividend yield D/P) and cross-sectional momentum for US stocks. It is observed that momentum profits exist with both 'cheap' and 'expensive' stocks but the difference between winners and losers is far greater in the latter group. Furthermore, a portfolio of expensive winners has historically returned considerably more than a portfolio of cheap losers. Little difference was observed in the returns of cheap winners and expensive winners. Ap Gwilym et al (2009) observe similar results for UK equities when looking at the relationship between dividend yield and momentum. They observe that the highest returns came from selecting winners from a group of zero-dividend stocks.

We find that the equally weighted index returns' portfolio shows significant downside negative skewness and substantial maximum drawdown over all periods and combinations that we examine. We also find that this negative skewness is reproduced in the case of relative momentum-only strategies. This feature of momentum returns was recently highlighted by Daniel and Moskowitz (2016) and Daniel et al (2012) who note that momentum strategy returns are often skewed and are subject to momentum crashes where momentum portfolio returns fall abruptly following a downturn in the market overall, the point being that momentum buying is based on a positive relative performance attribute, rather than an absolute one. This means that relative momentum buying continues in downturns, including severe downturns.

A number of different approaches to understanding and combating downside risk have been examined in the literature. Strub (2013) compares two alternative solutions for tail risk, especially in the left tail. These are options or cash-based methods. The latter solutions were originally examined in the context of portfolio insurance in the 1980's, (see Brennan and Schwartz, 1989, for example). Strub (2013) shows, using simulations of data from the last decade, that options-based management of tail risk using widely-available traded options, has a significantly higher cost than cash-based strategies which allocate a fraction of portfolios to cash depending on measured Value at Risk compared with a target. The cash-based strategies generate higher returns for a given reduced level of volatility than the standard options-based alternatives.

As Israelov (2017) states: *“For those who are concerned about their equity’s downside risk, reducing their equity position is significantly more effective than buying protection. Sized to achieve the same average return, divesting has lower drawdowns, lower volatility, lower equity beta, and a higher Sharpe ratio than does buying put options.”*

Regarding derivatives’ based protection in the case of equities, unless the options’ purchases and their maturities are timed precisely right around equity drawdowns of uncertain length, then they may result in little downside protection, and even make things worse by increasing rather than decreasing drawdowns and volatility per unit of expected return. This approach also echoes the views of Ilmanen (2016). He refers to index put buying as protection for equity portfolios as (looking at historical data) “roughly a minus one Sharpe strategy”. Of course, very fast bear markets and crashes can be protected against by using puts, but this is so expensive relative to the slow crash alternative (moving into cash) which is so successful in the long run. “Trend-following has a clear positive Sharpe ratio, and it has done well in most of the historical bear markets over the past hundred years”, Ilmanen (2016).

In this paper, we show that trend-following strategies offer similar improvements in terms of reduced downside risk and maximum drawdown to those shown by Strub (2013): simple intuition would suggest that buying downside protection when volatility rises is bound to be expensive, whereas moving into cash has no such pricing issues.

4. Data

Fundamentally, Value investing is based on the idea that cheaply priced stocks outperform more expensive stocks in the long term. It is seen by some as a “pro-cyclical” factor, outperforming during periods of economic expansion, suggesting that market timing may be possible. Researchers and investors differ on the best representation of ‘Value’, allowing a range of possible definitions in practice. As an example, the MSCI Enhanced Value Index applies three valuation ratio descriptors on a sector relative basis:

- Forward price to earnings (Fwd P/E);
- Enterprise value/operating cash flows (EV/CFO); and
- Price to book value (P/B).

The basic idea of having 3 criteria is that while the use of a single ratio may be inherently sensible, it is often the case that individual value stocks fall into a 'value trap' and prices do not rebound as expected: this can be avoided by using further criteria such as forward PE ratios.

This study utilizes MSCI Value and Growth indices for 23 developed and 21 emerging countries. As we have seen, characteristics for stocks being included in Value indices include book-to-market value and forward price-earnings ratios, whilst Growth attributes include historical and forward EPS growth rates and historical sales growth rates¹. Monthly data in US dollars for these total return indices begins from the end of December 1974 for developed markets and the end of December 1996 for emerging markets unless indicated otherwise in parentheses:

Developed

Australia, Austria, Belgium, Canada, Denmark, Finland (12/1987), France, Germany, Greece (12/1996), Hong Kong, Israel (12/1996), Italy, Japan, Netherlands, New Zealand (12/1987), Norway, Portugal (12/1996), Singapore, Spain, Sweden, Switzerland, United Kingdom and United States.

Emerging

Brazil, Chile, China, Colombia, Czech Republic, Egypt (09/1997), Hungary, India, Indonesia (05/1997), Korea, Malaysia (05/1997), Mexico, Peru, Philippines, Poland, Russia, South Africa, Taiwan, Thailand (05/1997), Turkey and UAE (05/2014).

5. Results

(a) Basic Value and Growth results

Table 1 shows the summary statistics for Value and Growth over the study period. The results are presented in two periods: 1976-2017, the whole data period, and 1998-2017, the period that emerging markets are introduced². All portfolios are equal weighted and rebalanced monthly. Firstly, we note the conventional relationship of Value outperforming Growth across

¹ See <http://www.msci.com/products/indices/style/vgvars.html> for a full description.

² We use the first year of data for momentum calculations in subsequent sections hence the somewhat later start than might be anticipated based on the data section.

all periods and for both developed and emerging markets. Returns are typically 2% higher per year for Value, (contrary, for example, to Houge and Loghran, 2006) whilst volatilities are similar for both strategies. We also observe the large outperformance of emerging markets over developed markets in the latter time period irrespective of whether one was invested in Value or Growth. Volatility was higher for emerging markets but the far greater levels of return more than compensated for this, with the Sharpe ratios all substantially greater for emerging assets than for developed ones: for example, at 0.46 for emerging Value versus 0.31 for developed Value. The difference is even greater for Growth portfolios at 0.39 versus 0.23. The full period (1976-2017) developed portfolio Sharpe ratios are far higher than for the shorter (1998-2017) period ratios, emphasising how relatively poor has been recent performance for developed equities. However a major consistent feature across all portfolios is the substantial drawdowns in all eras for all investing styles-at around 55-60% in all cases. Clearly such drawdowns are a major disadvantage in holding equity portfolios.

We next consider the interaction of relative momentum with Value and Growth. Following the method of Ilmanen (2011) we rank markets according to their prior 12-month return but volatility-weight these by dividing by the standard deviation of returns over the same period. Portfolios are held for 1-month and then momentum is recalculated. (Relative) Momentum portfolios include either the top five or top quarter of countries based on the rankings. We do this since the number of countries available varies over the data period and we can assess the sensitivity of our results to changes in country composition. The top five ensures a constant number of countries whereas the top quarter ensures a constant proportion.

Table 2 shows that relative momentum improves the risk-adjusted returns of developed markets (relative to Table 1) but not in the case of emerging markets. For the period 1976-2017, the Sharpe ratio for developed markets increases from around 0.37-0.48 (see Table 1) to around 0.56-0.68 (Table 2) with the addition of a momentum filter. The developed growth portfolio returns an average of over 17% in Table 2 but only 10.56% in Table 1: clearly 'hot' countries continues to offer excellent returns in subsequent months. There is also a similar magnitude improvement in the period of 1988-2017, with developed growth portfolio returns rising from 6.12% pa to 10.17% with the relative momentum selection, and a commensurate rise in Sharpe ratios from, say, 0.23 for developed growth to 0.45 for the relative momentum

equivalent. During this same time frame, relative momentum strategies outperform the base (no momentum, Table 1) case within emerging markets.

The far-right hand columns of Table 2 show results of when developed and emerging markets are combined in a single, relative momentum portfolio: here, relative momentum again improves performance quite noticeably. Whilst relative momentum may not have added very much benefit *within* emerging markets, it has when selecting *between* emerging and developed countries.

In terms of the relative momentum effect within Value and Growth, we observe that returns and volatilities remain fairly similar in developed Value portfolios irrespective of whether momentum is introduced. Emerging Value portfolios have slightly lower risk-adjusted returns with a relative momentum filter. The big gains, however, from relative momentum accrue in the Growth portfolios. For developed Growth portfolios, returns increase by around 7% per annum with only slight increases in volatility (Tables 1 and 2). The Sharpe ratios of these Growth portfolios are now higher than those of comparable Value portfolios. As noted earlier, the results are less impressive within emerging markets but again substantially higher returns are displayed when all developed and emerging countries are combined versus the base case in Table 1 (17.07% versus 12.55%). The evidence presented here thus supports the earlier findings of Asness (1997) and ap Gwilym et al (2009) that relative momentum exhibits a larger effect within Growth stocks than Value. However, note that the maximum drawdowns for all strategies, wherever relative momentum or simple base case, are *not* reduced: the strategies still have negatively skewed returns and large drawdowns which would most likely lead to destructive margin calls in practice (see Gray and Vogel, 2013).

(b) Factor Timing

Impact of Relative Momentum

Table 3 displays the results of using relative momentum in a different way. This time all countries are equally-weighted with a country's allocation being invested in either their Value or Growth index. The investment decision is based on whichever has the higher volatility-adjusted momentum with rebalancing taking place each month. This method makes it less likely that a few countries dominate with their performance (which could conceivably happen with say the Top 5 portfolios). Using this method we observe that for developed markets, risk-adjusted returns are slightly higher than the base cases of either just Value or Value and

Growth combined, but are lower than the momentum returns exhibited in Table 2. Emerging market returns are comparable to the base case in Table 1 and slightly higher than the relative momentum results in Table 2 (12.33% versus 12.72%). Negative skewness and large drawdowns are still a feature of these portfolios.

Impact of Introducing Absolute Momentum

Thus far, we have looked only at relative momentum and found some evidence of outperformance over standard portfolios. We now turn our attention to trend following (or absolute momentum) as a comparison. Consistent with Faber (2007) we use a 10-month moving average³ to define the trend, though the precise length of lag is not too important (see Clare et al, 2016). If the current price is above the simple average of the most recent 10 end of month prices then the asset is in an uptrend and is purchased (or held if already owned). If the current price is below the 10-month moving average (10MA) then the asset is sold and the proceeds invested in short-term treasury bills. Recalculation takes place at the end of each month.

Table 4 reports the results of trend following on Value and Growth portfolios with no relative momentum selection present. For the period 1976-2017, we note that returns are slightly lower for Value portfolios (12.44% pa) than without trend following (12.55%), slightly higher for Growth portfolios (11.17% versus 10.56%) and approximately the same for when the two types are combined. The major difference though is the volatilities. With trend following these are around 10.56%-10.70% annualized compared to around 16.51-16.78% without. This equates to Sharpe ratios being approximately 0.25 higher in the case of the former (ie trend adjusted). We also note the big difference in maximum drawdown experienced with the application of trend following. These are typically constrained to around 20% as opposed to 57%-61% when trend following is not used. The results are fairly similar regardless of whether Value, Growth or a combination of the two are held in portfolios.

³ Faber (2007) and Clare et al (2016) show that a variety of moving average lengths between 6-12 months perform similarly across a range of asset classes.

For the period 1998-2017 we again observe substantial outperformance for trend following versus the base case. Returns for portfolios containing developed markets are higher with emerging market returns fairly similar. Volatilities and drawdowns are again substantially reduced leading to considerably higher Sharpe ratios (around 0.68-0.74, (Table 4) versus 0.37-0.48 (Table 1)). With the addition of trend following, we observe that Value and Growth portfolios now exhibit rather less difference in returns (1.27%, (Table 4) versus 1.99% (Table 1)). Relative to the base case, trend following has made the largest difference to growth stocks, though.

Comparing the effect of trend following with relative momentum (Table 2), we find that risk-adjusted returns are higher and more consistent using the former, particularly for the latter period. Nominal returns in the period 1976-2017 were higher using relative momentum but these came at the expense of much increased volatility. It is really only in the developed Growth sector that relative momentum finds any outperformance over trend following. We also note that relative momentum suffers similar large drawdowns that buy-and-hold portfolios also exhibit. One of the most appealing characteristics of trend following, given that many investors are assumed to experience loss aversion, appears to be the ability to substantially *reduce* drawdowns by being out of the market during major bear markets. This supports previous evidence presented by Faber (2007) and Clare et al (2013). A final observation is that trend following portfolios display less negative skew than relative momentum equivalents, which again is perceived to be a more desirable quality from an investment perspective.

Dual Momentum: Combining Relative and Absolute Momentum

The most appealing characteristic of relative momentum we have noted thus far is the very high level of nominal return that it generates, albeit at the cost of high volatility. Trend following, by contrast, has demonstrated the ability to dampen volatility. Therefore, we next consider whether there is any benefit to combining relative momentum with trend following. Antonacci (2012) refers to this as 'dual' momentum.

Table 5 displays the results of combining the two approaches. Portfolios are formed in the same fashion using relative momentum as previously described in Table 2. In this instance

though, the risk asset is only included in the portfolio if it is also in a positive trend as described by the 10MA rule. Should an index pass the relative momentum filter but fail the trend following filter then this part of the portfolio allocation will be turned over to T-Bills. Thus indices must be both relative and absolute winners to be included. We find that the level of return is considerably higher than using trend following alone, at say 17.85% p.a. (top5 , Table 5) for developed Growth momentum portfolios versus 11.17% for trend following alone (see Tables 4 and 5), and the Sharpe ratio rising from 0.63 to 0.80. There are many instances where returns in Table 5 are greater than the comparables of just relative momentum shown in Table 2. Volatility, skewness, and drawdowns are also higher in Table 2, however, than pure trend following. This leads to Sharpe ratios that are typically not too dissimilar from those in Table 4 but higher than those in Table 2. The main beneficiaries of the dual approach are the developed Growth stocks. Portfolios containing these have Sharpe ratios in the 0.8 range for the full 1976-2012 period. The high nominal returns of these growth stocks also make them attractive to investors that are prepared to operate near the more extreme end of the risk-reward spectrum but may still suffer leverage aversion (see Asness, 1997).

The evidence presented here demonstrates the role that momentum, particularly the absolute kind, has as an investment technique. Both Value and Growth strategies benefit from it but the latter seems to gain more. From a practitioner's perspective, it seems particularly unwise to invest in Growth stocks without the tailwind of a positive trend. It may be that these stocks are harder to value, e.g. traditional value metrics are harder to use when a company has no earnings, or perhaps even no sales too, than firms that have historic track records of profitability. The dot-com boom perhaps provides illustration of the most recent difficulties that investors face when assigning a price to this type of security. As further support, Faber (2011) compares the performance of buy-and-hold versus trend following over 12 asset price bubbles during the last 100 years. The results show improved real returns to trend following with considerably less volatility and lower maximum drawdowns. Trend following thus can be a valuable reference point for investors when emotions are running high.

Transactions Costs

Further, as regards transactions' costs; while these will have varied very substantially over our sample period, and hence would be impossible to introduce with any degree of accuracy to

our calculations, we can certainly say that there is no evidence of over-trading with less than one round trip transaction per country per year on average over our sample period. In fact, for both developed and emerging, and for Value and Growth, the average only ranges between 0.83 and 0.87 transactions per year. This suggests that, whatever the charges, the relative returns would be unaffected. Further, ETFs and similar instruments are only recently being introduced for these countries and styles and hence any practical implementation would involve a swap being constructed by an investment bank. Where ETFs are available for these countries, the developed ones should cost less than 50bp per year and the emerging ones around 65bp, with transaction fixed charges of only a few bps per purchase or sale, and under one transaction per asset per year on average. These low fees will not adversely affect our conclusion that once the left tail risk is managed, Growth will outperform Value in both absolute and risk-adjusted terms, for both developed and emerging countries.

6. Risk-Adjusted Returns

The properties thus far refer to unconditional returns from trend following and momentum strategies, though we do, of course, discuss Sharpe ratios. In this section we examine whether these excess returns are explained by widely employed risk factors. For clarity, we examine the returns from particular strategies: these are Value only (Table 1), top-5 momentum Growth (Table 2), Value trend following (Table 4) and Growth momentum (Table 5). In particular, we examine the estimates of alphas after regressing the returns from the strategies on two sets of risk factors: the Fama-French four US equity market factors, MKT (the 'market'), SMB (size), HML (value), and UMD (momentum), (see Table 6), and secondly, a wider set market factors: the excess return from the GS Commodity Market index (GSCI), the MSCI world equity index (MSCI) and the Barclays Aggregate Bond Index (BAR). These are risk factors identified by Asness et al (2013) as significant in a range of markets. In Table 6 we see that the negative skewness of the Value portfolios is lower than Growth and that TF reduces skewness in both but more in Value than Growth. The Bai-Ng tests (see Bai and Ng, 2005, and Daniel and Moskowitz, 2016) say that the skewness is borderline significant for the two non-TF strategies. In terms of the levels of returns, all have significant alpha after exposing them to Fama-French and international risk factors. Some of the raw return is explained by these risk factors in each case. Growth returns are always higher even after the risk factor adjustment -

maybe this is due to some extra return being compensation for crash or liquidity risk (as shown by the amount of skewness). Clare et al (2014) show that the significant negative skewness in returns from momentum strategies in European stock markets can be partly explained by large negative shocks to measures of funding liquidity. By contrast, Asness et al (2013) show that Value portfolio returns are negatively related to funding liquidity risk, a more extreme version of the relatively reduced negative skewness that we find in Value returns.

7. Conclusion

We have investigated the relationship between Value and Momentum across a wide range of international equity markets. A distinction is made as well between cross-sectional, or relative, momentum where assets are ranked according to their performance against each other, and trend following or absolute momentum, where assets are categorized according to whether they have recently exhibited positive nominal return characteristics.

It is observed that on a buy-and-hold basis Value has outperformed Growth but this gap diminishes when a relative momentum filter is applied. Indeed, for portfolios containing developed markets, Growth actually displays higher risk-adjusted returns. The exception is within emerging markets where Value still outperforms. When trend following is applied to Value and Growth portfolios both show substantially improved risk-adjusted returns compared to the base case scenario. Volatilities are reduced by around 40% and maximum drawdowns decline markedly too. Compared to relative momentum, trend following delivers a more favourable investment performance, particularly in the case of emerging markets. The exception is in developed Growth stocks where relative momentum still delivers small outperformance. Table 7 shows annual returns with and without trend following, emphasising how removing the large drawdowns for emerging markets in 2000 and both developed and emerging in 2008 has such a beneficial effect on performance.

Finally, we consider the combination of relative momentum and trend following. We find that whilst Sharpe ratios were similar in many cases to just trend following alone, the nominal level of return is considerably higher. Volatility is also commensurately higher and maximum

drawdowns increase too. These remain more subdued though than when relative momentum is applied alone. The benefits accrue to both Value and Growth portfolios but particularly to the latter.

We thus suggest that both Value and Growth strategies can be enhanced using momentum. A trend following overlay offers particularly substantial benefits compared to a buy-and-hold approach. Growth stocks especially seem suited to such a method, possibly as a result of them being harder to value in some instances. Trend following provides an unemotional guide to the prevailing market conditions.

Smart beta strategies are being developed to reflect the many additional factors that research has identified as being associated with higher returns. Further analysis of the potential benefits from rotating between additional factors, momentum and trend following provides an obvious research agenda following the results in this paper.

8. References

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Table 1									
Summary Statistics									
	Developed			Emerging			All Countries		
	Value	Growth	All	Value	Growth	All	Value	Growth	All
<i>1976-2017</i>									
Annualized Return (%)	12.55	10.56	11.59	-	-	-	13.76	11.77	12.55
Annualized Volatility (%)	16.78	16.51	16.44	-	-	-	17.26	17.32	17.20
Sharpe Ratio	0.48	0.37	0.43	-	-	-	0.54	0.42	0.47
Max. Monthly Return (%)	18.53	13.98	15.20	-	-	-	18.50	15.50	16.64
Min. Monthly Return (%)	-25.32	-23.99	-24.66	-	-	-	-26.89	-27.34	-27.11
Maximum Drawdown (%)	61.25	57.51	59.40	-	-	-	59.19	59.37	59.27
Skew	-0.80	-0.90	-0.88	-	-	-	-0.80	-0.91	-0.88
<i>1998-2017</i>									
Annualized Return (%)	7.95	6.12	7.09	12.73	10.99	11.89	10.39	8.57	9.51
Annualized Volatility (%)	19.36	18.30	18.57	23.30	23.28	23.15	20.25	19.84	19.90
Sharpe Ratio	0.31	0.23	0.28	0.46	0.39	0.43	0.42	0.34	0.38
Max. Monthly Return (%)	18.53	13.98	15.20	19.02	19.02	18.29	18.50	15.50	16.64
Min. Monthly Return (%)	-25.32	-23.99	-24.66	-28.69	-31.18	-29.94	-26.89	-27.34	-27.11
Maximum Drawdown (%)	61.25	57.51	59.40	56.82	61.59	59.25	59.19	59.37	59.27
Skew	-0.68	-0.77	-0.76	-0.61	-0.77	-0.71	-0.69	-0.80	-0.76

Table 2									
Relative Momentum									
	Developed			Emerging			All Countries		
	Value	Growth	All	Value	Growth	All	Value	Growth	All
<i>TOP 5: 1976-2017</i>									
Annualized Return (%)	14.37	17.23	15.50	-	-	-	15.37	19.01	17.07
Annualized Volatility (%)	17.67	18.70	18.30	-	-	-	19.35	20.00	20.53
Sharpe Ratio	0.56	0.68	0.60	-	-	-	0.56	0.73	0.61
Max. Monthly Return (%)	15.80	15.83	16.90	-	-	-	21.76	18.35	24.23
Min. Monthly Return (%)	-34.27	-34.65	-34.27	-	-	-	-34.27	-34.65	-35.57
Maximum Drawdown (%)	61.02	51.39	63.49	-	-	-	62.12	59.25	62.92
Skew	-1.26	-1.00	-1.14	-	-	-	-0.86	-0.99	-0.85
<i>TOP 5: 1998-2017</i>									
Annualized Return (%)	9.40	10.17	11.68	12.63	9.53	12.72	11.42	13.72	12.79
Annualized Volatility (%)	18.23	18.46	19.51	26.33	25.91	20.68	21.55	21.20	22.17
Sharpe Ratio	0.41	0.45	0.50	0.41	0.29	0.52	0.44	0.56	0.49
Max. Monthly Return (%)	13.80	15.83	16.90	22.94	17.73	17.60	21.76	18.35	24.23
Min. Monthly Return (%)	-25.09	-20.00	-28.70	-35.68	-34.84	-31.56	-30.76	-29.63	-29.10
Maximum Drawdown (%)	61.02	51.39	63.49	62.24	67.90	60.61	62.12	59.25	62.92
Skew	-0.78	-0.42	-0.63	-0.68	-0.98	-0.87	-0.29	-0.54	-0.31

Table 2 (continued)									
Relative Momentum									
	Developed			Emerging			All Countries		
	Value	Growth	All	Value	Growth	All	Value	Growth	All
<i>TOP QUARTER: 1976-2017</i>									
Annualized Return (%)	14.67	17.49	16.03	-	-	-	15.50	18.51	17.29
Annualized Volatility (%)	17.59	18.83	17.63	-	-	-	18.30	19.75	18.54
Sharpe Ratio	0.58	0.69	0.65	-	-	-	0.60	0.71	0.69
Max. Monthly Return (%)	15.80	18.23	15.05	-	-	-	16.90	17.60	15.61
Min. Monthly Return (%)	-34.27	-34.65	-34.16	-	-	-	-34.27	-34.65	-34.16
Maximum Drawdown (%)	59.27	50.48	57.51	-	-	-	63.49	60.61	60.99
Skew	-1.30	-0.99	-1.30	-	-	-	-1.14	-1.19	-1.24
<i>TOP QUARTER: 1998-2017</i>									
Annualized Return (%)	10.01	10.70	9.91	12.50	9.54	11.38	11.68	12.72	12.43
Annualized Volatility (%)	18.07	18.75	17.75	26.52	25.87	24.88	19.51	20.68	19.66
Sharpe Ratio	0.45	0.47	0.45	0.40	0.30	0.38	0.50	0.52	0.54
Max. Monthly Return (%)	12.89	18.23	15.05	22.94	17.73	20.08	16.90	17.60	15.61
Min. Monthly Return (%)	-25.91	-23.20	-24.56	-35.68	-34.84	-35.48	-28.70	-31.56	-30.13
Maximum Drawdown (%)	59.27	50.48	57.51	62.24	67.90	64.14	63.49	60.61	60.99
Skew	-0.84	-0.42	-0.86	-0.72	-0.99	-0.89	-0.63	-0.87	-0.80

Table 3			
Switching Between Value and Growth for Each Country			
	Developed	Emerging	All
<i>1976-2017</i>			
Annualized Return (%)	13.21	-	14.21
Annualized Volatility (%)	16.26	-	17.04
Sharpe Ratio	0.54	-	0.57
Max. Monthly Return (%)	15.22	-	16.48
Min. Monthly Return (%)	-23.64	-	-26.79
Maximum Drawdown (%)	57.40	-	58.12
Skew	-0.85	-	-0.89
<i>1998-2017</i>			
Annualized Return (%)	8.66	12.33	10.67
Annualized Volatility (%)	18.21	23.21	19.68
Sharpe Ratio	0.37	0.45	0.45
Max. Monthly Return (%)	15.22	19.39	16.48
Min. Monthly Return (%)	-23.64	-30.42	-26.79
Maximum Drawdown (%)	57.40	59.09	58.12
Skew	-0.75	-0.74	-0.80

Table 4									
Trend Following									
	Developed			Emerging			All Countries		
	Value	Growth	All	Value	Growth	All	Value	Growth	All
<i>1976-2017</i>									
Annualized Return (%)	12.44	11.17	11.83	-	-	-	13.15	12.03	12.61
Annualized Volatility (%)	10.70	10.56	10.42	-	-	-	10.91	10.92	10.75
Sharpe Ratio	0.74	0.63	0.70	-	-	-	0.79	0.69	0.75
Max. Monthly Return (%)	11.32	10.30	10.06	-	-	-	12.17	13.32	11.15
Min. Monthly Return (%)	-18.53	-20.58	-19.55	-	-	-	-18.53	-20.58	-19.55
Maximum Drawdown (%)	18.66	20.75	19.71	-	-	-	18.66	20.75	19.71
Skew	-0.63	-0.95	-0.84	-	-	-	-0.41	-0.77	-0.63
<i>1998-2017</i>									
Annualized Return (%)	9.60	8.57	9.12	12.33	12.02	12.20	11.07	10.34	10.72
Annualized Volatility (%)	10.46	9.85	9.89	13.44	13.13	13.11	10.94	10.66	10.64
Sharpe Ratio	0.74	0.68	0.73	0.78	0.77	0.79	0.84	0.79	0.83
Max. Monthly Return (%)	11.32	10.30	9.92	15.95	16.78	14.46	12.17	13.32	11.15
Min. Monthly Return (%)	-10.79	-9.06	-9.93	-11.07	-11.17	-11.12	-9.21	-9.19	-9.12
Maximum Drawdown (%)	17.24	18.04	17.63	18.82	18.43	17.03	16.46	16.27	16.36
Skew	-0.10	-0.15	-0.19	0.59	0.30	0.43	0.29	0.13	0.17

Table 5									
Combination of Trend Following and Relative Momentum									
	Developed			Emerging			All Countries		
	Value	Growth	All	Value	Growth	All	Value	Growth	All
<i>TOP 5: 1976-2017</i>									
Annualized Return (%)	14.46	17.85	16.58	-	-	-	15.94	19.75	17.65
Annualized Volatility (%)	15.46	16.60	17.11	-	-	-	17.07	17.83	18.91
Sharpe Ratio	0.64	0.80	0.71	-	-	-	0.67	0.86	0.70
Max. Monthly Return (%)	15.80	15.83	22.10	-	-	-	21.76	18.35	24.23
Min. Monthly Return (%)	-34.27	-34.65	-35.57	-	-	-	-34.27	-34.65	-35.57
Maximum Drawdown (%)	34.27	35.21	35.57	-	-	-	40.96	38.38	40.04
Skew	-1.24	-1.01	-1.00	-	-	-	-0.65	-0.79	-0.68
<i>TOP 5: 1998-2017</i>									
Annualized Return (%)	10.11	12.46	12.89	14.19	12.87	13.38	13.11	16.32	15.08
Annualized Volatility (%)	14.46	15.04	15.30	20.62	20.16	23.49	17.92	17.82	19.28
Sharpe Ratio	0.57	0.70	0.72	0.60	0.54	0.49	0.63	0.81	0.68
Max. Monthly Return (%)	13.80	15.83	22.10	20.87	17.73	24.52	21.76	18.35	24.23
Min. Monthly Return (%)	-15.51	-16.73	-15.96	-28.81	-27.92	-36.39	-15.51	-16.73	-15.96
Maximum Drawdown (%)	28.62	35.21	27.15	41.32	47.37	48.01	40.96	38.38	40.04
Skew	-0.30	0.10	0.18	-0.16	-0.35	-0.29	0.40	0.18	0.27

Table 5 (continued)									
Combination of Trend Following and Relative Momentum									
	Developed			Emerging			All Countries		
	Value	Growth	All	Value	Growth	All	Value	Growth	All
<i>TOP QUARTER: 1976-2017</i>									
Annualized Return (%)	14.61	18.08	16.30	-	-	-	16.30	19.15	17.74
Annualized Volatility (%)	15.29	16.63	15.42	-	-	-	15.83	16.86	15.99
Sharpe Ratio	0.66	0.82	0.76	-	-	-	0.75	0.87	0.83
Max. Monthly Return (%)	15.80	18.23	15.05	-	-	-	15.80	17.60	15.61
Min. Monthly Return (%)	-34.27	-34.65	-34.16	-	-	-	-34.27	-34.65	-34.16
Maximum Drawdown (%)	34.27	34.65	34.16	-	-	-	34.27	35.48	34.16
Skew	-1.28	-0.96	-1.27	-	-	-	-0.97	-0.99	-1.06
<i>TOP QUARTER: 1998-2017</i>									
Annualized Return (%)	10.40	12.94	11.42	14.05	12.91	12.80	13.85	15.08	14.33
Annualized Volatility (%)	14.08	15.11	13.88	20.69	20.15	20.08	15.34	15.68	15.23
Sharpe Ratio	0.60	0.73	0.69	0.59	0.55	0.54	0.78	0.84	0.82
Max. Monthly Return (%)	12.63	18.23	15.05	20.87	17.73	17.22	14.04	17.60	15.61
Min. Monthly Return (%)	-14.38	-16.88	-16.18	-28.81	-27.92	-32.07	-14.59	-15.60	-14.69
Maximum Drawdown (%)	25.95	34.09	28.16	42.49	47.37	44.30	24.05	35.48	32.62
Skew	-0.34	0.23	-0.20	-0.19	-0.35	-0.55	0.21	0.07	0.11

Table 6: Alpha calculations for a selection of investment strategies (1977 Jan to 2017 Dec)

This table presents the unconditional mean returns (column 4, panel A) “Average”, generated by the different investment strategies: Panel A also reports the results of regressing the returns from these strategies using Fama and French (1992) three factors, MKT, SMB and HML, plus Carhart’s (1997) momentum factor, UMD. Panel B reports the results of regressing the returns from these strategies against a set of wider risk the Goldman –Sachs Commodity Index (GSCI), the world equity market return index (MSCI), Barclays Bond Index (BAR). Newey and West (1997) t-statistics are shown in square brackets. Prob F is based on an F-statistic for the test of the joint significance of the independent variables. π is the Bai-Ng (2005) test for skewness.

Panel A	Skew	π	Average	Alpha	MKT	SMB	HML	UMD	Prob F
Value Only	-0.797	-1.85	1.22 [4.70]	0.713 [3.80]	0.830 [15.04]	0.089 [1.46]	0.220 [3.70]	-0.161 [3.81]	0
GrowthT5Mom	-1.185	-1.86	1.61 [5.31]	0.930 [3.88]	0.882 [10.62]	0.106 [1.27]	0.122 [1.24]	0.0921 [1.28]	0
Value TF	-0.415	-0.903	1.09 [6.40]	0.734 [4.73]	0.426 [8.70]	0.0555 [1.25]	0.139 [2.25]	0.0630 [1.98]	0
Growth Mom TF	-0.988	-1.23	1.61 [6.71]	1.027 [4.80]	0.640 [7.82]	0.110 [1.36]	0.0769 [0.81]	0.215 [3.49]	0
Panel B					Alpha	GSCI	MSCI	BAR	Prob F
Value Only					0.620 [4.59]	0.00704 [4.32]	0.0851 [24.35]	-0.00608 [0.87]	0
GrowthT5Mom					1.00 [5.06]	0.00729 [2.13]	0.0884 [15.11]	-0.00290 [0.27]	0
Value TF					0.793 [6.24]	0.00223 [0.99]	0.0436 [9.82]	-0.00284 [0.50]	0
Growth Mom TF					1.15 [5.78]	0.00247 [0.72]	0.0650 [8.63]	-0.00793 [0.74]	0

Table 7								
Annual Returns of Value and Growth With and Without Trend Following								
Year	Value				Growth			
	Developed		Emerging		Developed		Emerging	
	No TF	With TF	No TF	With TF	No TF	With TF	No TF	With TF
1976	8.5	8.0			1.7	2.4		
1977	5.0	9.7			1.0	5.5		
1978	31.4	27.9			22.2	18.3		
1979	19.4	16.7			31.2	28.5		
1980	15.2	15.5			22.7	17.0		
1981	0.6	9.7			-6.7	7.6		
1982	-2.1	5.8			-7.6	6.8		
1983	34.1	29.5			23.8	19.7		
1984	4.0	2.0			-1.9	1.4		
1985	63.1	58.6			66.6	57.0		
1986	46.9	40.9			52.6	49.2		
1987	11.4	11.8			5.4	5.3		
1988	26.2	16.1			22.8	14.2		
1989	30.0	23.1			26.8	19.7		
1990	-8.9	-5.2			-9.8	-2.3		
1991	15.1	4.0			14.3	1.8		
1992	-6.1	-1.0			-4.1	-0.7		
1993	48.9	34.6			34.1	21.5		
1994	5.8	-0.2			8.3	3.5		
1995	17.6	13.2			20.3	15.1		
1996	20.6	17.5			18.4	15.6		
1997	14.0	12.8			14.0	10.7		
1998	17.1	14.2	-16.1	-11.2	25.8	20.5	-15.2	-10.4
1999	26.9	19.8	66.8	39.8	31.4	31.2	64.0	37.1
2000	-2.4	-4.4	-26.0	-1.7	-19.8	-5.4	-30.8	-6.7
2001	-12.8	-7.7	5.1	-7.2	-21.4	-2.1	-5.9	-1.1
2002	-12.1	-5.7	1.9	-0.1	-13.4	-6.7	8.3	2.1
2003	52.2	45.2	72.3	56.5	40.9	33.1	66.2	55.7
2004	32.8	30.7	43.4	36.1	24.2	19.6	36.7	27.0
2005	16.2	15.2	41.1	36.2	12.8	11.0	38.6	33.8
2006	33.8	33.3	38.4	30.7	29.5	28.5	39.1	29.8
2007	13.3	14.1	44.6	43.9	26.3	25.1	42.4	40.6
2008	-49.0	-7.5	-44.7	-12.1	-47.9	-10.7	-51.5	-10.4
2009	50.4	27.9	75.6	46.2	41.6	21.5	71.6	34.8
2010	4.0	-9.0	24.7	16.0	12.0	2.3	25.9	15.8
2011	-17.6	-2.1	-16.1	-8.7	-16.6	-3.1	-17.9	-7.9
2012	21.0	4.6	25.5	4.3	18.6	2.6	21.4	3.8
2013	27.7	22.1	-9.8	-5.6	17.5	11.0	-4.2	-0.7
2014	-3.7	3.4	-2.2	-1.0	-6.8	-0.8	-0.4	3.2
2015	-10.5	-3.8	-20.2	-2.8	-0.3	-3.2	-12.9	-1.2
2016	6.7	1.9	18.8	2.6	-1.6	-4.8	12.0	2.4
2017	25.0	21.0	29.0	21.8	27.9	19.3	32.3	21.4

